

LazySlide: accessible and interoperable whole-slide image analysis

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Histopathological data are foundational in both biological research and clinical diagnostics but remain siloed from modern multimodal and single-cell frameworks. Here we introduce LazySlide, an open-source Python package built on the scverse ecosystem for efficient whole-slide image analysis and multimodal integration. By leveraging vision–language foundation models and adhering to scverse data standards, LazySlide bridges histopathology with omics workflows. It supports tissue and cell segmentation, feature extraction, cross-modal querying and zero-shot classification, with minimal setup.

Histopathology is a cornerstone of biomedical research and clinical practice, offering high-resolution insights into the spatial organization of tissues, cellular morphology and pathological alterations. Advances in digital pathology have enabled the collection of large repositories of whole-slide images (WSIs), creating unprecedented opportunities to study the tissue basis of human health and disease. However, computational analysis of WSIs remains challenging. Existing toolkits for WSI analysis, such as QuPath¹, CLAM², PathML³, Slideflow⁴ and TIAToolbox⁵, offer partial solutions but are often limited by fragmented data structures, platform-specific constraints and high technical barriers to entry. These limitations hinder integration with the multimodal and single-cell workflows that are increasingly standard in modern biology.

To address these limitations, we developed LazySlide, a general-purpose Python framework for WSI analysis, fully interoperable with the scverse ecosystem⁶ widely used in biomedical research. LazySlide streamlines WSI preprocessing, segmentation, feature extraction and multimodal integration using state-of-the-art deep learning models—effectively bridging histopathology with omics workflows, data structures and operations.

At its core, LazySlide introduces a custom data structure, WSIData (Fig. 1), built upon scverse's SpatialData⁷ but optimized for the diverse formats and scale of WSIs. While SpatialData supports only OME-TIFF pyramids, most WSIs are stored in proprietary formats requiring specialized readers. Existing extensions such as SOPA⁸ adapt SpatialData to WSIs but incur substantial disk overhead due to serialization (5–10× increase). By contrast, WSIData enables efficient, direct access to standard WSI formats without duplication, while maintaining compatibility with scverse tools and deep learning pipelines.

The framework follows familiar application programming interface (API) conventions from AnnData⁹, Scanpy¹⁰ and Squidpy¹¹, reducing the barrier for computational pathologists and genomics researchers alike.

LazySlide includes a full suite of tools for WSI analysis, from preprocessing, quantification, visualization and multi-omics integration to advanced capabilities that support insightful interpretation and interactive exploration, leveraging zero-shot approaches and genomic data integration (Fig. 1). Specifically, slides are segmented into tissue regions, tiled into memory-efficient patches and embedded via vision foundation models into high-dimensional feature spaces. These embeddings support downstream analyses such as unsupervised clustering, which can reveal spatial tissue architectures—for example, the mucosa, submucosa, muscularis and lymphoid tissue in the human small intestine (Fig. 1). Built-in cell segmentation and classification tools enable the quantification of cellular composition and morphology.

Beyond image-based analysis, LazySlide leverages vision–language models to support natural language queries, enabling users to localize image regions most similar to prompts, such as ‘lymphocyte’. These features also enable multimodal integration with transcriptomics and other data types, supporting disease scoring and mechanistic interpretation. LazySlide further provides zero-shot capabilities for slide captioning, relevance scoring and text-guided segmentation, as well as efficient, declarative visualization of WSIs and annotations, including segmented regions, cells and extracted features.

LazySlide supports a diverse range of applications and facilitates rigorous benchmarking against existing methodologies, thereby showcasing its robust capabilities and broad utility. To showcase its

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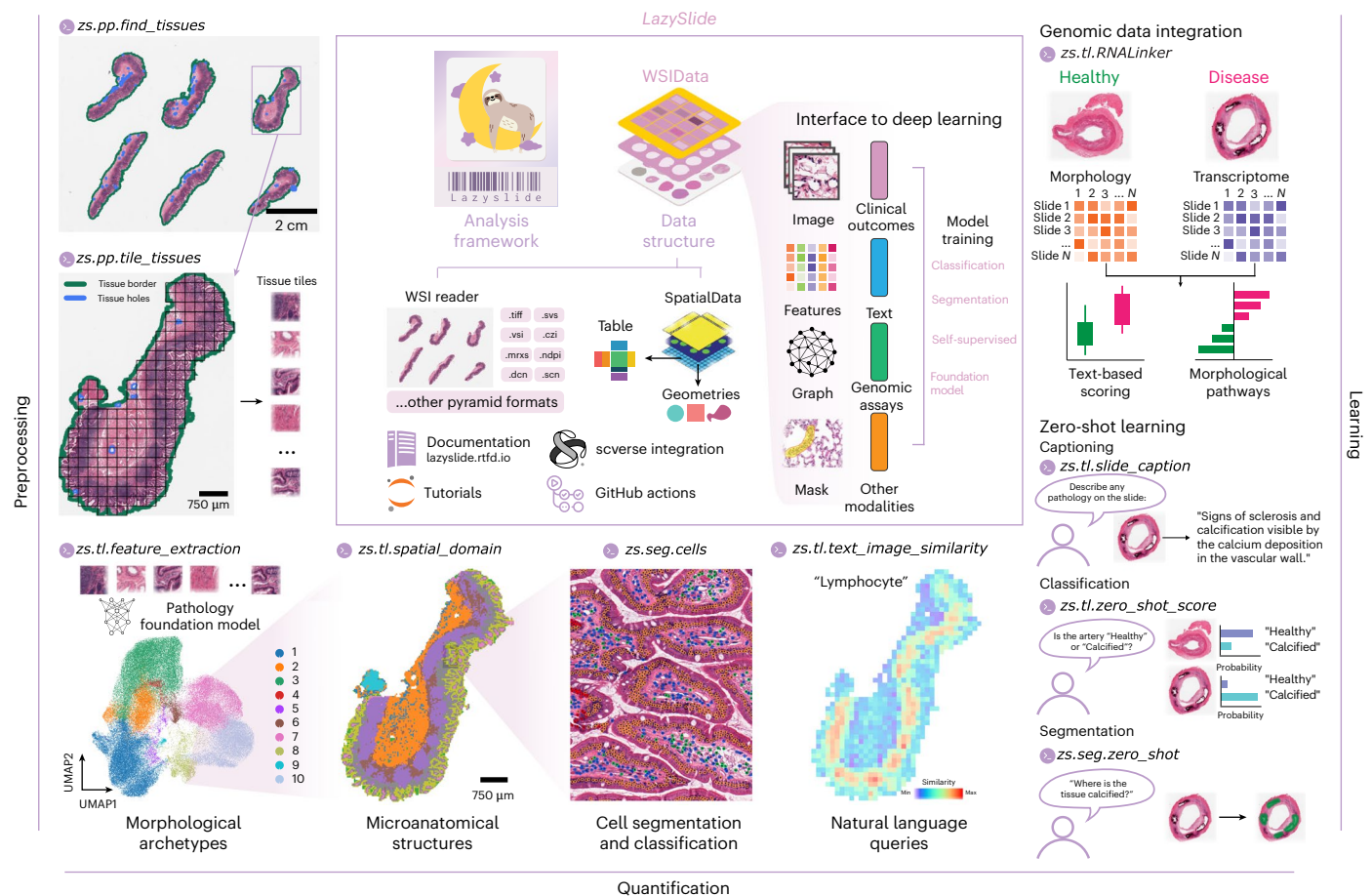


Fig. 1 | LazySlide framework overview. WSIData structure enables WSI preprocessing, quantification of cellular, morphological and microanatomical tissue features, and multimodal integration with deep learning frameworks. scverse logo reproduced from ref. 7 under a Creative Commons license [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/).

capabilities in combining WSI data with other modalities, we demonstrate three representative applications: (1) zero-shot vision–language querying, (2) multimodal integration of histology and transcriptomics and (3) zero-shot organ classification—each illustrating cutting-edge use cases for computational pathology and omics data integration.

First, we assembled a dataset of human artery slides from the GTEx project¹², including healthy ($n = 24$) and calcified ($n = 21$) tissues with matched RNA sequencing (RNA-seq) profiles (Fig. 2a). With only five lines of code, users can compute text-to-image similarity maps across each slide. Terms related to ‘calcification’ show higher enrichment in calcified samples, whereas anatomical terms predominate in healthy tissues (Fig. 2b,c). A differential analysis of image–text features highlights terms such as gap junction, vascular niche and apoptosis as significantly enriched in calcified arteries, consistent with observed morphological changes. Building on this, a slide-level ‘calcification score’ is computed using top- k pooling over similarity maps with the term calcification (Fig. 2d), yielding significantly elevated scores in calcified tissues.

To explore joint imaging–transcriptomic patterns, we use the calcification score to anchor a multimodal integration of WSI-derived and RNA-seq embeddings (Fig. 2e). This analysis, which is executable in a single line of code, shows that image features separate healthy and calcified groups more distinctly in Uniform Manifold Approximation and Projection (UMAP) space than RNA-seq alone. This pattern is also captured via multi-omics factor analysis (MOFA)¹³ integration using Muon¹⁴, the scverse multi-omics framework (Fig. 2f). We then compared gene-level relevance with calcification by ranking genes from RNA-only differential expression or joint WSI + RNA integration. Pathway enrichment on the top 300 up- and downregulated genes

from each method revealed that only the WSI and RNA integration identifies key calcification-related pathways, including IL-18 signaling (Fig. 2g), and yields a greater number of significant pathway associations overall (Fig. 2h).

Finally, we demonstrate the zero-shot classification capabilities of LazySlide. Using vision–language models, WSIs from nine distinct human organs were queried against their organ names (Fig. 2i). With a single line of code, LazySlide correctly identified the majority of organ sources (Fig. 2j). Applying the same approach to the artery calcification dataset also yields high classification performance (Fig. 2k), which illustrates the ability to extract valuable insights without tissue- or context-specific training.

To assess usability and performance, we benchmarked LazySlide against established tools across a standard preprocessing pipeline: tissue segmentation, tiling, tile dataset preparation for PyTorch¹⁵ and feature extraction. LazySlide completes this workflow with fewer lines of code, lower token count and a simpler API, facilitating rapid development and code maintenance (Fig. 2l). We also compared LazySlide with QuPath in a classification setting of semantically labeled microanatomical domains of murine breast cancer lung metastasis using extracted features by different vision models. Excluding ResNet50, which is not trained on tissue images, LazySlide consistently outperforms QuPath-derived features (Fig. 2m). Moreover, tissue segmentation is markedly faster in LazySlide compared with QuPath using either automated or manual workflows (Fig. 2n). Finally, we performed a qualitative comparison of LazySlide’s features with existing software (Extended Data Table 1), highlighting its remarkable flexibility and the number of unique tasks it supports, including text–image

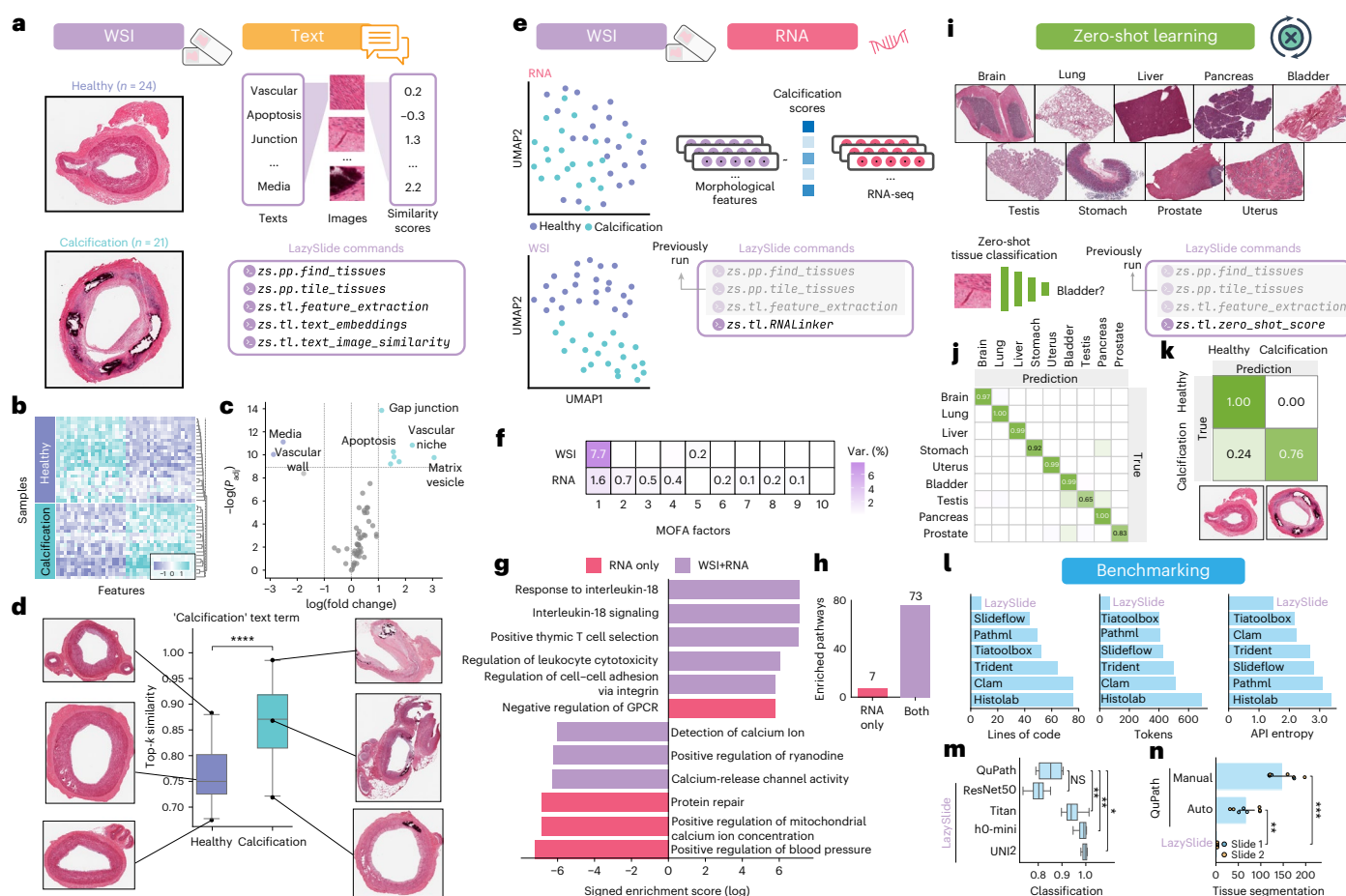


Fig. 2 | LazySlide applications and benchmarks. a, WSIs of healthy and calcified arteries. **b**, Text term scores across WSIs showing calcification-associated patterns. **c**, Differential text features between conditions (two-sided Mann–Whitney U test, P_{adj} is calculated with the Bonferroni method). **d**, Calcification scores per WSI with representative tiles. **** $P < 0.0001$ ($P = 0.00001$, two-sided Mann–Whitney U test). **e**, UMAP of RNA-seq (top) and WSI features (bottom) from paired samples. **f**, Variance explained by MOFA of RNA and WSI modalities. **g**, Enriched pathways from WSI + RNA versus RNA-only analysis. **h**, Count of highly enriched pathways by analysis type. **i**, Representative tissues for zero-shot

classification. **j**, Confusion matrix of zero-shot organ classification. **k**, Zero-shot calcification prediction probabilities. **l**, Code complexity comparison across frameworks. **m**, Classification accuracy versus QuPath. NS, non-significant; * $P < 0.05$; ** $P < 0.01$ (unpaired two-sided Student's t -test; QuPath versus ResNet50 $P = 0.22$, versus Titan $P = 0.044$, versus h0-mini $P = 0.0059$, versus UN12 $P = 0.0028$). **n**, Tissue segmentation time versus QuPath. ** $P < 0.01$; *** $P < 0.001$ (two-sided Mann–Whitney U test; LazySlide versus QuPath-auto $P = 0.0024$, versus QuPath-manual $P = 0.0007$). Box plots: medians, quartiles, 1.5 $\times$ interquartile range. Error bars: mean $\pm$ 95% confidence interval.

queries, RNA data integration, zero-shot learning, unsupervised spatial domain detection, declarative visualization and virtual staining.

Despite its advantages, LazySlide's strict use of the Zarr format maximizes interoperability but its distributed file structure may limit flexibility in compute environments with file count restrictions. Future developments will focus on co-registration capabilities across imaging modalities and for three-dimensional tissue reconstruction.

Altogether, LazySlide represents a substantial advancement in artificial-intelligence-enabled histopathology and tissue biology, bridging computational pathology with multimodal omics through a modular, user-friendly and open-source framework. By supporting a wide range of foundational models and ensuring seamless interoperability with the scverse ecosystem, LazySlide enables integrative, scalable and interpretable analysis of WSIs. It empowers both computational pathologists and genomics researchers to uncover new insights into tissue biology and disease, accelerating the development of data-driven, clinically meaningful models.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information,

acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41592-026-03044-7>.

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Methods

Data structure

The WSIData object was designed as an inheritance of SpatialData, allowing it to behave exactly like SpatialData⁷. This design enables users familiar with scverse and SpatialData to readily use WSIData across scverse packages such as LazySlide. The SpatialData component of WSIData is a data container, storing all analysis metadata and results generated within LazySlide. An additional layer and corresponding APIs were implemented to specifically interact with WSIs. WSIData was built to support multiple WSI backends, including OpenSlide¹⁶, tiffslide¹⁷, bioformats¹⁸ and cuCIM¹⁹, ensuring compatibility with a broad range of image formats. An accessor interface, similar to those found in pandas²⁰, xarray²¹ and SpatialData, was also implemented to provide a variety of features. For instance, users can easily retrieve useful information, such as the number of tissue pieces found in each slide, the number of tiles in a tessellated image, and an AnnData object with morphological features through a ‘fetch’ accessor. The ‘iter’ accessor allows users to iterate through each tissue piece, tile or their features. The ‘dataset’ accessor provides an interface to create various PyTorch datasets that could be directly used for training or running deep learning models.

Tissue segmentation

Tissue segmentation was performed using a multistep image processing pipeline, which was implemented with OpenCV. The algorithm initiates processing of WSIs at an automatically determined optimal resolution level. For artifact filtering, specialized thresholding techniques were applied to identify and exclude nontissue regions on the basis of color properties. Alternatively, images can be either converted to grayscale or transformed to HSV (Hue, Saturation, Value) color space, with the saturation channel extracted to enhance tissue visibility. A median blur filter was first applied to reduce noise, followed by binary thresholding, using either Otsu’s method or a fixed threshold value. The resulting binary mask undergoes morphological closing operations to fill small gaps within tissue regions while preserving overall tissue structure. This process effectively separated tissue regions from background and artifacts, enabling subsequent region identification and analysis. Tissue masks can be further refined using higher-resolution image data when specified, and regions below a configurable minimum area threshold are excluded. As an alternative, the GrandQC²² pretrained tissue segmentation model was also integrated into LazySlide to perform deep-learning-based tissue segmentation.

Tissue tiling

Tissue regions were tiled using an automated approach that generated a grid of tiles within identified tissue masks or contours. The algorithm allows users to request arbitrary tile sizes no lower than the raw image resolution and accepted user-specified tile dimensions along with customizable stride or overlap between adjacent tiles. For each tissue region, tiles were initially generated within its bounding box, then filtered to retain only those intersecting with the tissue contour. To eliminate tiles containing excessive background, a configurable background fraction threshold is applied, discarding tiles where the background exceeded this threshold. The tiling process supports multiresolution analysis by allowing the user to specify a resolution in micrometers per pixel, ensuring a consistent physical scale across slides scanned at different magnifications.

Quality control

Following tiling, quality control was implemented through a configurable scoring system that evaluated tiles on the basis of metrics including focus quality, contrast, brightness and tissue characteristics. This scoring facilitated subsequent filtering of low-quality tiles containing artifacts or out-of-focus regions, thereby ensuring the reliability of downstream analysis. All operations supported parallel processing

for computational efficiency when handling large datasets. In addition, GrandQC²² was incorporated in LazySlide to detect common artifacts such as bubbles, tissue folds and out-of-focus regions.

Feature extraction and aggregation

The feature extraction module supports a variety of pretrained vision models through Torch Image Models (timm, <https://github.com/huggingface/pytorch-image-models>). The tiled images are processed through selected models, including classic vision architectures pretrained on ImageNet (for example ResNet, DenseNet and ViT) and pathology-specific foundation models (for example UNI/UNI²³, Virchow/Virchow2^{24,25}, H-Optimus-0/1 and GigaPath²⁶), to extract high-dimensional feature representations. Feature extraction supports automatic device selection (central processing unit/graphics processing unit), mixed-precision inference and configurable batch processing for computational efficiency. For each tissue tile, the deep learning model extracts feature vectors that capture complex visual patterns not easily quantifiable through traditional image analysis. Following extraction, LazySlide implements multiple feature aggregation strategies to generate slide-level or tissue-level representations from tile-level features. These include statistical aggregation methods (mean and median) and specialized neural encoders that incorporate spatial information (for example, PRISM²⁷ and TITAN²⁸).

Natural language query of tissue images

Natural language query capabilities in LazySlide leverage multimodal foundation models to bridge vision and language domains, enabling content retrieval through text-based queries. The framework uses state-of-the-art pathology-specific models, such as PLIP²⁹ and CONCH³⁰, to facilitate semantic search within WSIs. The query process follows a two-stage approach: first, text queries are transformed into dense vector representations using the text encoder component of the chosen model. Second, these text embeddings are compared with pre-extracted image feature vectors using cosine similarity to identify regions most relevant to the query. Similarity scores are computed via the dot product between normalized text and image embeddings, with higher scores indicating stronger semantic correspondence. This powerful functionality enables users to search for specific histological patterns, cell types or tissue structures using natural language descriptions, offering a remarkable advantage over traditional visual inspection methods.

Spatial domain detection

LazySlide implements unsupervised spatial domain identification through a multistage, graph-based clustering approach. Initially, a neighborhood graph is constructed from the extracted tiles, where each tile represents a node and edges connect spatially adjacent tiles. Optionally, morphological features can be smoothed by incorporating neighboring information, thereby enhancing feature consistency. Following graph construction, LazySlide uses a dimensionality reduction pipeline, as introduced in Scanpy, which first scales the feature data and then applies principal component analysis to capture dominant variation patterns. The resulting low-dimensional representation is then used to construct a weighted *k*-nearest neighbor graph. Finally, the Leiden community detection algorithm³¹ (implemented via igraph) identifies coherent tissue domains by partitioning this graph into communities, with the clustering resolution controlled by a user-adjustable parameter. This comprehensive approach effectively segments histologically distinct regions within tissue slides without requiring manual annotation, thereby enabling automated identification of intricate tissue architectural features.

Whole-slide cell segmentation and classification

LazySlide supports cell segmentation of WSIs through Instanseg³² and Cellpose³³. Joint cell segmentation with classification through

Nulite³⁴ and HistoPLUS³⁵. The segmentation stages operate within a dedicated SegmentationRunner class, which efficiently handles batch processing across tiles, utilizing graphics processing unit acceleration when available, and performs the crucial merging of results from individual tiles. For each tiled image, cells are identified and their contours are transformed into a polygon representation. To merge results across tiles of the WSI, results are aggregated by addressing potential boundary artifacts, via a sophisticated polygon merging algorithm that leverages a spatial indexing tree (STR tree). All initially overlapping polygons are identified and then merged by preserving their class labels while calculating probability-weighted averages for cells spanning multiple tiles, thereby ensuring spatial continuity across the entire tissue section. Newly merged polygons are then pruned from the tree, and the process iterates until the tree contains no remaining branches, ensuring all overlapping polygons have been successfully merged into cohesive cell instances.

Integration with RNA-seq data

LazySlide enables multimodal data integration through the RNALinker class, which links histopathological features with paired transcriptomic profiles. Morphological features extracted from WSIs are first aggregated into slide-level representations that are matched to corresponding genomic samples. The integration proceeds in two steps: (1) samples are grouped and scored on the basis of morphological features, either through differential analysis between defined sample groups or by using morphologically derived annotations, such as text-based scores; (2) these morphology-based scores are statistically associated with gene expression data using correlation metrics (for example, Pearson, Spearman and Kendall) and regression models (for example, linear regression and lasso). This approach identifies genes whose expression is strongly associated with specific histological patterns. By linking morphology to transcriptomic signatures, this integration supports hypothesis generation about the molecular mechanisms driving tissue phenotypes and facilitates the discovery of biomarkers that bridge computer vision-derived features with biological interpretation.

Zero-shot learning

LazySlide supports zero-shot learning through multimodal vision-language foundation models such as PRISM²⁷ and TITAN²⁸, supporting various biologically relevant questions without task-specific training. For zero-shot classification, slide-level feature embeddings are compared with arbitrary, user-defined text prompts, yielding probability scores that quantify the semantic alignment between tissue morphology and natural language descriptions. Alternatively, for descriptive analysis, LazySlide generates natural language summaries of histological content from slide embeddings and text prompts, enabling automated preliminary reporting and assisting pathologists with concise, context-aware interpretations. For zero-shot segmentation of tissue, LazySlide uses text-image similarity metrics to produce a binary mask based on a user-defined threshold. Detected object bounding boxes from this mask are then used to guide segmentation via the Segment Anything Model 2 (SAM2)³⁶, enabling flexible and label-free identification of tissue regions of interest.

Benchmarking

To assess the usability and interoperability of LazySlide within the scverse ecosystem, we conducted a benchmark comparison against several widely used WSI processing libraries: CLAM², TRIDENT³⁷, PathML³, Tiatoolbox⁵, Histolab³⁸ and Slideflow⁴. Each library was evaluated using a standardized digital pathology workflow comprising tissue segmentation and tiling, feature extraction with ResNet50, construction of a PyTorch dataset for tile-level access (to assess ease of integration with deep learning), and generation of an AnnData object containing tile features and spatial metadata (to assess compatibility

with scverse workflows). Benchmark scripts were implemented for each library, formatted using ruff, and executed in isolated Docker environments to ensure consistency and reproducibility. To quantify implementation complexity, we measured token counts, lines of code and API entropy, providing a comparative view of conceptual and practical overhead.

For direct benchmarking against QuPath, we used four mouse lung tissue slides from patient-derived xenograft models of breast cancer with metastasis. These slides were previously semantically annotated for three tissue classes: tumor, airways and blood vessels. Feature extraction was performed using both QuPath and LazySlide (with ResNet50, h0-mini³⁹, UNI2²³ and TITAN²⁸ backbones), and classification accuracy was used for performance comparison. Tissue segmentation was benchmarked by comparing QuPath (manual and automated methods) with LazySlide. Following a standardized protocol, four experienced volunteers used QuPath's manual labeling and thresholding tools, and the time needed to perform tissue segmentation was recorded for two slides. The time performance of LazySlide was measured using the find_tissues function.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

GTEX WSIs and RNA expression data are available from the GTEX portal (<https://gtexportal.org>). WSIs of patient-derived xenograft models of breast cancer and their semantic segmentation are available via Zenodo at <https://doi.org/10.5281/zenodo.15497223> (ref. 40).

Code availability

LazySlide and WSIData are publicly available via GitHub at <https://github.com/RendeiroLab/LazySlide> and <https://github.com/RendeiroLab/WSIData>, published on the Python package index (PyPI) and conda-forge, available via Zenodo at <https://doi.org/10.5281/zenodo.14939805> (ref. 41) and <https://doi.org/10.5281/zenodo.15083425> (ref. 42), and documented at <https://lazyslide.readthedocs.io> and <https://wsidata.readthedocs.io>. A companion repository on benchmarking is also publicly available via GitHub at <https://github.com/RendeiroLab/LazySlide-benchmark>.

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Author contributions

Y.Z. and A.F.R. conceptualized the study; Y.Z. developed LazySlide with contributions from E.A.; E.C. and I.B. provided annotations for WSIs; J.W. and A.F.R. supervised the research. All authors reviewed and approved the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Extended Data Table 1 | Software feature comparison

Feature	LazySlide	CLAM	Trident	TIAToolbox	Slideflow	PathML	histolab
PyData/scverse ecosystem interoperability	✓	X	X	X	X	X	X
Tissue segmentation + Tiling	✓	✓	✓	✓	✓	✓	✓
Feature extraction	✓	✓	✓	✓	✓	X	X
Text-image query	✓	X	X	X	X	X	X
Integration with RNA-seq	✓	X	X	X	X	X	X
Cell segmentation	✓	X	X	✓	✓	X	X
Semantic segmentation	✓	X	X	✓	X	X	X
Artifact detection	✓	X	✓	X	X	X	X
Zero-shot classification	✓	X	X	X	X	X	X
Slide captioning	✓	X	X	X	X	X	X
Unsupervised spatial domain detection	✓	X	X	X	X	X	X
PyTorch datasets for deep learning	✓	✓*	X	✓	✓	✓	X
Declarative visualization	✓	X	X	X	X	X	X
Virtual staining	✓	X	X	X	X	X	X
Model training	X	✓*	X	X	✓	✓	X
Image registration	X	X	X	✓	X	X	X
Graphic user interface	X	X	X	X	✓	X	X
	✓ Available	X Non-available				*Only for multiple instance learning	

Comparison of features between LazySlide and existing software.

Reporting Summary

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Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☒	☐ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	No original data were collected in this study.
Data analysis	Packages developed in this work: lazyslide v0.6.0, wslide v0.6.1 Direct dependencies: cyclopts v3.14.2, legendkit v0.3.4, matplotlib v3.10.1, matplotlib-scalebar v0.9.0, psutil v6.1.1, rich v14.0.0, scikit-learn v1.6.1, seaborn v0.13.2, timm v1.0.15, torch v2.7.0, Dependency group "all": scanpy v1.10.4, scipy v1.15.2, torchstain v1.4.1, torchvision v0.22.0, transformers v4.51.3 Dependency group "dev": albumentations v2.0.6, jupyterlab v4.4.1, matplotlib v3.10.1, matplotlib-scalebar v0.9.0, pre-commit v4.2.0, pytest v8.3.5, ruff v0.11.8, scanpy v1.10.4, scikit-learn v1.6.1, scipy v1.15.2, scjvava v1.12.0, segmentation-models-pytorch v0.5.0, spatialdata-plot v0.2.10, taskipy v1.14.1, torchstain v1.4.1, torchvision v0.22.0 Dependency group "docs": myst-nb v1.2.0, numpdydoc v1.8.0, pydata-sphinx-theme v0.16.1, sphinx v8.2.3, sphinx-book-theme v1.1.3, sphinx-contributors v0.2.7, sphinx-copybutton v0.5.2, sphinx-design v0.6.1 Dependency group "model": conch v0.1.0, einops v0.8.1, einops-exts v0.0.4, environs v14.1.1, sacremoses v0.1.1, transformers v4.51.3, Dependency group "tutorials": bokeh v3.7.2, dask-jobqueue v0.9.0, gseapy v1.1.8, igraph v0.11.8, ipywidgets v8.1.6, marsilea v0.5.2, matplotlib-venn v1.1.2, mofapy2 v0.7.2, mpl-fontkit v0.5.1, muon v0.1.7, parse v1.20.2, pypalettes v0.1.5

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
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GTEX whole slide images and RNA expression data are available from its portal (<https://gtexportal.org>). Whole slide images of a breast cancer PDX models and their semantic segmentation are available at Zenodo (<https://doi.org/10.5281/zenodo.15497223>).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\)](#), [and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	Sex representation in GTEx samples used is representative of the donor population which is unbalanced in a 2:1 male:female ratio.
Reporting on race, ethnicity, or other socially relevant groupings	Race and ethnicity representation in GTEx samples used is representative of the donor population which is unbalanced, predominantly consisting of individuals of European ancestry.
Population characteristics	Age and other demographic and clinical characteristics are representative of the donor population in GTEx.
Recruitment	Not applicable.
Ethics oversight	No ethical approval necessary.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

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Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	No statistical determinations were done to determine sample sizes.
Data exclusions	Not applicable.
Replication	Not applicable.
Randomization	Not applicable.
Blinding	Not applicable.

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We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

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n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

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Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

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Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.